Andreea Elena Bodea

andreea.e.bodea@gmail.com | LinkedIn | Personal Website | GitHub

Passionate AI engineer specializing in natural language processing, machine learning, and privacy-preserving techniques. Experienced in delivering research-driven solutions and effectively managing complex projects. Committed to bridging theoretical insights with impactful, real-world AI applications.

Education

- **Master of Science in Computer Science** 10/2022-09/2025
  Technical University of Munich
  - Master's Thesis: "Privacy Issues and Privacy Preserving Mechanisms in Retrieval-Augmented Generation (RAG) systems" (highest grade)
  - Major: Machine Learning and Natural Language Processing (NLP)
  - Lectures: Machine Learning, Deep Learning, Computer Vision
  - **Erasmus Exchange** at **Sorbonne University Paris** 09/2023-02/2024
- **Bachelor of Science in Business Administration** 10/2019-09/2023
  Ludwig Maximilian University of Munich
  - Bachelor's Thesis: "From freedom of speech to the U.S. Capitol riots: Topic evolution of content on Parler."
- **Bachelor of Science in Computer Science with Minor in Economics**
  Ludwig Maximilian University of Munich 10/2018-11/2022
  - Bachelor's Thesis: "Domain adapted topic modelling for alt-tech social media."

Work Experience

- **AI Researcher** 10/2024-05/2026
  Department of Computer Science (TUM) in partnership with the **Federal Ministry for Economic Cooperation and Development (BMZ)**
  - Authored a review research paper on privacy in RAG systems by systematizing literature and creating a widely applicable taxonomy of all privacy issues and privacy mitigation strategies
  - Developed an open-source application enhancing text privacy through advanced anonymization techniques and differential privacy at multiple RAG pipeline stages
- **Startup Project Co-Founder** 10/2024-01/2025
  AI Entrepreneurship Lab - AI startup incubator by TUM.ai
  - Conducted extensive market analysis by interviewing over 30 lawyers, identifying crucial pain points in German legal tech
  - Implemented an MVP for an AI-based search engine for legal cases, used by 5 lawyers, built with a LangChain Python backend and React frontend
  - Pitched the startup idea to an audience of 400 people, including a committee of investors
- **AI Engineer** 02/2024-09/2024
  Digital Product School by UnternehmerTUM in partnership with the **Federal Ministry of Economic Cooperation and Development of Germany (BMZ)** and the **GovTech Campus Germany**
  - Built an AI-powered dashboard enabling ministry employees to analyze unstructured PDF reports with an agentic RAG system using LlamaIndex, Pinecone, Azure, FastAPI, and Docker
  - Co-developed intuitive internal web application with TypeScript backend, React frontend, and PostgreSQL, significantly improving internal workflows
- **Teaching Assistant** for the **Software Development Practical Course: Java Programming** at the **Ludwig Maximilian University of Munich** 05/2022-08/2022
  - Facilitated agile project development as a Scrum Master, leading a team of 4 students to deliver a Java-based game project by coaching team members, introducing Agile engineering practices, and teaching Scrum principles
- **Teaching Assistant** for the **Software Engineering Lecture** at the **Ludwig Maximilian University of Munich** 10/2021-03/2022
  - Increased grading efficiency and fairness by evaluating assignments and exams for 100+ students, ensuring timely, constructive feedback

Technical Skills

- **Programming:**
  - Python
  - Java
  - SQL
  - TypeScript
- **LLM & RAG Tools:**
  - LlamaIndex
  - LangChain
  - LLM SDKs e.g. OpenAI
- **ML/NLP Libraries:**
  - Transformers
  - Scikit-Learn
  - SpaCy
- **Frameworks/Tools:**
  - PostgreSQL
  - Streamlit
  - React
  - FastAPI
  - Docker
- **Version Control/Others:**
  - GIT (GitHub, GitLab)
  - Jira, Miro

Soft Skills

- Leadership
- Agile practices
- Cross-functional teamwork
- Analytical thinking
- Time management
- Growth mindset

Languages

- **German C1**
  - Goethe-Zertifikat C1
  - Test DaF (grade 4.5)
- **English C1**
  - Cambridge Level 2 Certificate (Advanced)
- **French B1**
- **Romanian C2** - native

Interests

- IBM female mentoring
- Neuroscience research
- Language learning (Duolingo master)